NORMAL CROSSINGS DIVISORS CHARACTERIZED BY THE UNRAMIFIED NORMALIZATION AND REGULAR FIBRE PRODUCTS

THE STACKS PROJECT AUTHORS

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Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

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Source reference. Original reference 00NU: sources/stacks-chow/tags.txt, source lines 872–872. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a locally Noetherian scheme. Let $D \subset X$ be a closed subscheme. The following are equivalent*

(1) *D is a normal crossings divisor in X,*
(2) *D is reduced, the normalization $\nu : D^\nu \to D$ is unramified, and for any $n \geq 1$ the scheme*

$$Z_n = D^\nu \times_D \ldots \times_D D^\nu \setminus \{(p_1, \ldots, p_n) \mid p_i = p_j \text{ for some } i \neq j\}$$

is regular, the morphism $Z_n \to X$ is a local complete intersection morphism whose conormal sheaf is locally free of rank n.

Proof. First we explain how to think about condition (2). The diagonal of an unramified morphism is open (Morphisms, Lemma 02GE). On the other hand $D^\nu \to D$ is separated, hence the diagonal $D^\nu \to D^\nu \times_D D^\nu$ is closed. Thus Z_n is an open and closed subscheme of $D^\nu \times_D \ldots \times_D D^\nu$. On the other hand, $Z_n \to X$ is unramified as it is the composition

$$Z_n \to D^\nu \times_D \ldots \times_D D^\nu \to \ldots \to D^\nu \times_D D^\nu \to D^\nu \to D \to X$$

and each of the arrows is unramified. Since an unramified morphism is formally unramified (More on Morphisms, Lemma 02HE) we have a conormal sheaf $\mathcal{C}_n = \mathcal{C}_{Z_n/X}$ of $Z_n \to X$, see More on Morphisms, Definition 04F4.

Formation of normalization commutes with étale localization by More on Morphisms, Lemma 07TD. Checking that local rings are regular, or that a morphism is unramified, or that a morphism is a local complete intersection or that a morphism is unramified and has a conormal sheaf which is locally free of a given rank, may be done étale locally (see More on Algebra, Lemma 0AH0, Descent, Lemma 02VM, More on Morphisms, Lemma 069N and Descent, Lemma 05B2).

By the remark of the preceding paragraph and the definition of normal crossings divisor it suffices to prove that a strict normal crossings divisor $D = \bigcup_{i \in I} D_i$ satisfies (2). In this case $D^\nu = \coprod D_i$ and $D^\nu \to D$ is unramified (being unramified is local on the source and $D_i \to D$ is a closed immersion which is unramified). Similarly, $Z_1 = D^\nu \to X$ is a local complete intersection morphism because we may check this locally on the source and each morphism $D_i \to X$ is a regular immersion

as it is the inclusion of a Cartier divisor (see Lemma 0BIA and More on Morphisms, Lemma 069L). Since an effective Cartier divisor has an invertible conormal sheaf, we conclude that the requirement on the conormal sheaf is satisfied. Similarly, the scheme Z_n for $n \geq 2$ is the disjoint union of the schemes $D_J = \bigcap_{j \in J} D_j$ where $J \subset I$ runs over the subsets of order n. Since $D_J \to X$ is a regular immersion of codimension n (by the definition of strict normal crossings and the fact that we may check this on stalks by Divisors, Lemma 063I) it follows in the same manner that $Z_n \to X$ has the required properties. Some details omitted.

Assume (2). Let $p \in D$. Since $D^\nu \to D$ is unramified, it is finite (by Morphisms, Lemma 01WJ). Hence $D^\nu \to X$ is finite unramified. By Lemma 04HJ and étale localization (permissible by the discussion in the second paragraph and the definition of normal crossings divisors) we reduce to the case where $D^\nu = \coprod_{i \in I} D_i$ with I finite and $D_i \to U$ a closed immersion. After shrinking X if necessary, we may assume $p \in D_i$ for all $i \in I$. The condition that $Z_1 = D^\nu \to X$ is an unramified local complete intersection morphism with conormal sheaf locally free of rank 1 implies that $D_i \subset X$ is an effective Cartier divisor, see More on Morphisms, Lemma 069G and Divisors, Lemma 063L. To finish the proof we may assume $X = \mathrm{Spec}(A)$ is affine and $D_i = V(f_i)$ with $f_i \in A$ a nonzerodivisor. If $I = \{1, \ldots, r\}$, then $p \in Z_r = V(f_1, \ldots, f_r)$. The same reference as above implies that $(f_1, \ldots, f_r)$ is a Koszul regular ideal in A. Since the conormal sheaf has rank r, we see that $f_1, \ldots, f_r$ is a minimal set of generators of the ideal defining Z_r in $\mathcal{O}_{X,p}$. This implies that $f_1, \ldots, f_r$ is a regular sequence in $\mathcal{O}_{X,p}$ such that $\mathcal{O}_{X,p}/(f_1, \ldots, f_r)$ is regular. Thus we conclude by Algebra, Lemma 00NU that $f_1, \ldots, f_r$ can be extended to a regular system of parameters in $\mathcal{O}_{X,p}$ and this finishes the proof. $\square$